DHRUV DUBEY

AI Engineer | Agentic AI, LLM Systems & RAG | College Park, MD
+1 227-275-1372 | ddubey12@umd.edu | linkedin.com/in/dhruv-dubey-502a42284 | github.com/dhruvdubey3123

EDUCATION

University of Maryland, College Park — M.S., Data Science (GPA 4.00/4.00) *Aug 2025 – May 2027*

MIT-ADT University — B.Tech, Computer Science & Engineering (GPA 3.33/4.00) *2021 – 2025*

TECHNICAL SKILLS

Languages: Python, SQL, C++, R

Agents & LLMs: Agentic AI, Multi-Agent Systems, LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), Tool Calling, Prompt & Context Engineering, RAG, Vector Databases, Embeddings, LoRA/QLoRA Fine-Tuning, Snowflake Cortex

Evaluation: LLM & Agent Evaluation, Golden Datasets, LLM-as-Judge, RAGAS, Faithfulness & Context Metrics, CI Eval Gates, OpenTelemetry

Backend & MLOps: FastAPI, REST APIs, Pydantic, Microservices, Docker, Kubernetes (HPA, Probes), GitHub Actions CI/CD, MLflow, AWS, SLURM/HPC

ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, NLP, Computer Vision, Time Series, Anomaly Detection

Data: Snowflake, PostgreSQL, SQLite, Redis, MongoDB, Neo4j, Supabase, Power BI, Pandas, NumPy

EXPERIENCE

Allegis Global Solutions — AI Automation Engineer Intern *Jun 2026 – Sep 2026*

- Built a multi-agent triage system on Snowflake Cortex (Agents, Search, Analyst) that traces pipeline failures through a dependency graph to the broken stored procedure; root-cause diagnosis ~65% faster (weeks to days).
- Engineered semantic views and a YAML instruction layer built from prior tickets so the agent handles new failure patterns without retraining.
- Deployed it into analysts' Power BI workflow as a chat interface, cutting fix-proposal time from hours to minutes.

University of Maryland — Research Intern, Neuro-Symbolic AI *Sep 2025 – Sep 2026*

- Co-developed a six-stage reproducibility audit of neuro-symbolic AI: screened 5,497 records to 1,304 studies and reproduced 85 codebases on the Zaratan HPC cluster (SLURM, Docker).
- Found inconsistencies in setups, hyperparameters and evaluation methods across reasoning frameworks; co-authored the paper, now under review at IEEE TNNLS.

Fortytwo Labs — Machine Learning Intern *Feb 2025 – Jul 2025*

- Built TensorFlow autoencoders on login behavior (+40% anomaly detection accuracy) and KMeans/DBSCAN clustering that surfaced 35% more outliers.
- Integrated the models with the Drools rule engine for automated risk scoring, cutting manual security reviews 33%.

Intrainz — Data Science Intern *May 2023 – Jul 2023*

- Built a crop recommendation system on 5,000+ records, raising decision-support precision 15% across 150 pilot farmers.

PROJECTS

Pipeline Incident Triage Graph *github.com/dhruvdubey3123/pipeline-triage-graph*

LangGraph, Model Context Protocol, FastAPI, Pydantic, Docker, Kubernetes, GitHub Actions, pytest

- Open-source multi-agent triage: a tracer walks the pipeline DAG upstream from the symptom, then specialist and reviewer agents draft and check a fix behind a human-approval gate.
- CI gated on two 100-incident eval sets: root-cause accuracy 1.00; noisy logs drop category accuracy to 0.81, which the gate now tracks. Ships an MCP server and Kubernetes manifests.

Agent Harness Runtime *github.com/dhruvdubey3123/agent-harness-runtime*

Python, FastAPI, SQLite, Model Context Protocol, Pydantic, Docker, Kubernetes, GitHub Actions

- Agent runtime that owns the loop, context budget and per-agent tool sandbox, and logs every tool call so a crashed run resumes from its log instead of restarting.
- 200-run crash benchmark: 0 duplicated side effects vs 145 for restart-from-scratch; 143 runs recovered unattended vs 55.

Krishi Mitra — Multilingual Agricultural RAG Assistant

- Fine-tuned LLaMA 3.2 3B with RAG, answering farmers in Hindi, Marathi and English; piloted with 300 farmers, +20% query-resolution accuracy.

Smart Plant Health Monitor

- YOLOv5, ResNet and LSTM system classifying plant health at 90% accuracy, served as a Dockerized FastAPI microservice.

PUBLICATIONS

- Neuro-Symbolic AI Reproducibility Review — University of Maryland; arXiv:2608.26236; under review, IEEE Transactions on Neural Networks and Learning Systems.
- Krishi Mitra: LLM-Driven QA System for Indian Farmers — MIT-ADT University.